

File permissions in Linux

Project description

[Used Linux commands to manage file permissions. Used Linux commands to manage directory permissions.]

Check file and directory details

```
cd projects to get to the projects directory
ls -la lists the permissions of all files and directories in the
/home/researcher2/projects directory including hidden files.
project_k.txt -rw-rw-rw
project_m.txt -rw-r---
project_r.txt -rw-rw-r-
project_t.txt -rw-rw-r-
.project_x.txt -rw-w---
drafts drwx-x--
```

Describe the permissions string

[-rw-rw-rw string means that the user, group, and other only have reading and writing permissions to the `project_k.txt` file. - in the beginning represents that it's a file and not a directory, r represents reading, w represents writing, and - represents the lack of execute permissions.]

Change file permissions

[`project_k.txt` needs to have its permission modified so that others have no write permissions to any files. `Chmod o-w project_k.txt` will be used to modify the permissions and remove the write permission from others. `chmod` stands for change mode.]

```
project_k.txt -rw-rw-r-
```

Change file permissions on a hidden file

[`.project_x.txt` shouldn't have any write permissions for anyone, the user and group should only have read permissions. `chmod u=r,g=r .project_x.txt` will be used to

modify the permissions and overwrite the existing permissions to read only for the user and the group.]

```
.project_x.txt -r-r---
```

Change directory permissions

[`drafts` and its contents should only be accessed by the user. `chmod g-x drafts` will be used to modify these permissions and remove the execute permission from the group.]

```
drafts drwx---
```

Summary

[I navigated to the `projects` directory and checked the files and directory details. I noticed the least privilege principle wasn't in place and modified the files and directories to prevent security risks. Part of my job was to ensure that users on the research team are authorized with the appropriate permissions. I had to determine if the permissions matched the authorization given, and when they didn't I made modifications to permissions to authorize the appropriate users and remove any unauthorized access.]